

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A right circular cylinder has a volume of **432** cubic centimeters. The area of the base of the cylinder is **24** square centimeters. What is the height, in centimeters, of the cylinder?

Answer

- A. 18
- B. **24**
- C. 216
- D. 10,368

Correct Answer: A

Rationale

Choice A is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where πr^2 is the area of the base of the cylinder and h is the height. It's given that a right circular cylinder has a volume of **432** cubic centimeters and the area of the base is **24** square centimeters. Substituting **432** for V and **24** for πr^2 in the formula $V = \pi r^2 h$ yields **432 = 24h**. Dividing both sides of this equation by **24** yields **18 = h**. Therefore, the height of the cylinder, in centimeters, is **18**.

Choice B is incorrect. This is the area of the base, in square centimeters, not the height, in centimeters, of the cylinder.

Choice C is incorrect. This is the height, in centimeters, of a cylinder if its volume is **432** cubic centimeters and the area of its base is **2**, not **24**, cubic centimeters.

Choice D is incorrect. This is the height, in centimeters, of a cylinder if its volume is **432** cubic centimeters and the area of its base is $\frac{1}{24}$, not **24**, cubic centimeters.